Interface-Agnostic Contrastive & Energy-Balance Training Procedure for Monomer Crystallization

Self-supervised 376 × 376 polar partition function + local slide/twist perturbations + universal SE(3) involution vs. open-generator link function

a Training Datasets, Monomer Feature Graph & Two-Stage Loss Functions

Dataset 1: PDB Corpus & Monomer Graph

- 20 PDB Structures: 125 symmetry interfaces (H2O, Cl-)
- Monomer (193L): N = 1001 heavy atoms, P = 376 polar
- 91D 3D Env. Descriptors $\mathbf{f}_i \rightarrow$ Laplacian L_{chem}

Dataset 2: Positive Crystal Contacts (8 Poses)

- 7 Unique $P4_32_12$ Lattice Modes (m0 … m6) + capture
- Primary Partner Filter: $u_{ij} \geq 0.25$ or $\geq 0.70\, u_{i,\max}$
- Support-Balanced Weight: $w_{\text{norm}}(m) = 200/\max(\Psi_m, 1)$
- Universal support weighting (no hand-tuned weights)

Dataset 3: Perturbations & Held-Out Decoys

- 225 Local Slide/Twist Poses around each crystal x_m: $\delta t_\parallel \in \{4.5, 7.0\}$ Å, $\delta\theta \in \{-24^\circ, 0^\circ, +24^\circ\}$
- 61 Held-Out MC Decoys: never seen in training, used strictly for out-of-sample validation (61/61 < 0)

Stage A: Contrastive Embedding Loss ($\mathbf{u}_i, \mathbf{v}_i \in S^7$)

$s_{ij} = \frac{1}{2}(\mathbf{u}_i \cdot \mathbf{v}_j + \mathbf{v}_i \cdot \mathbf{u}_j)$ on $P \times P = 376 \times 376$ matrix

$$\mathcal{L}_{\text{Stage A}}(\mathbf{U}, \mathbf{V}) = -\sum_{i,j} W_{ij}^{\text{pos}} \ln \frac{\exp(s_{ij}/\tau)}{\sum_k \pi_{ik} \exp(s_{ik}/\tau)} + \frac{1}{2} \sum_{i,k \notin \mathcal{P}} W_{ik}^{\text{neg}} [s_{ik} + 0.52]_+^2 + 5 \sum_{i,j \in \mathcal{P}} W_{ij}^{\text{pos}} [0.85 - s_{ij}]_+^2 + \lambda_{\text{lap}} \operatorname{Tr}(\mathbf{U}^\top L_{\text{chem}} \mathbf{U} + \mathbf{V}^\top L_{\text{chem}} \mathbf{V}) \quad (\tau = 0.15)$$

Prior: $W_{ik}^{\text{neg}} = (16\, I_{\text{act}} + 1.2\, I_{\text{bg}} + W_{ik}^{\text{pert}})\, I_{(i,k) \notin \mathcal{P}}$

Stage B: Energy-Balance Activity Loss ($a_i \in [0.20, 1.65]$)

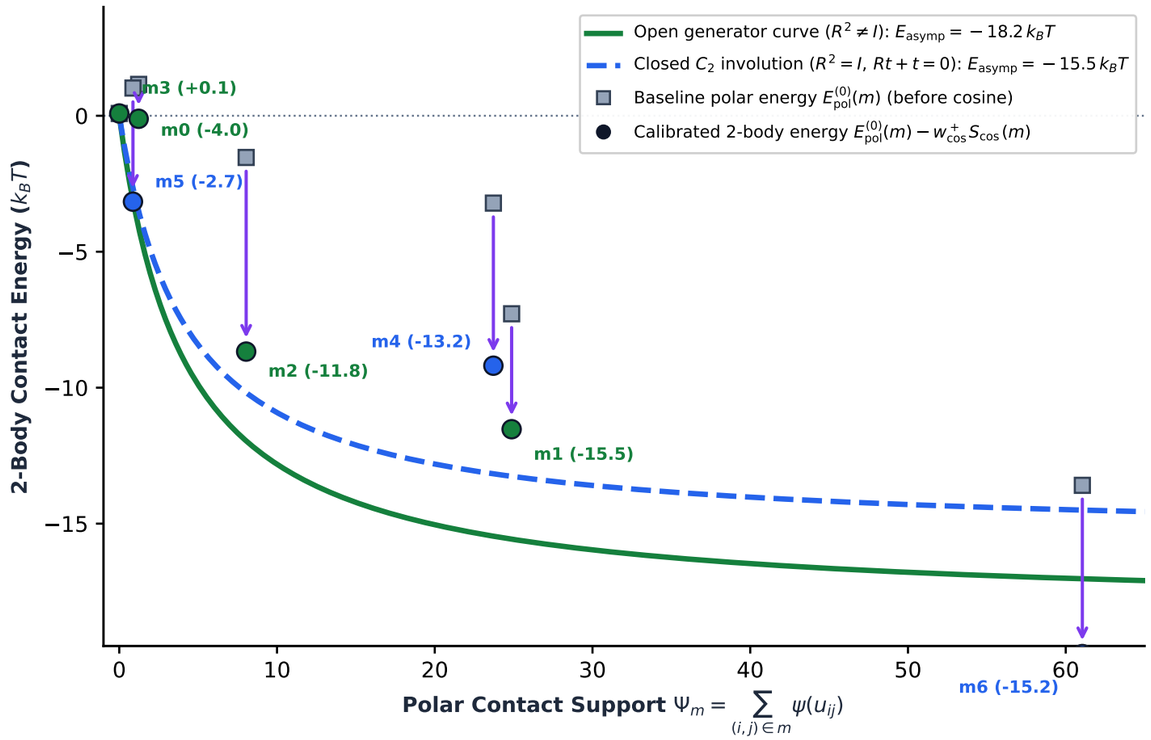
Universal Target Link: $S_{\cos}^*(m) = \max\left(0.55,\ \frac{E_{\text{pol}}^{(0)}(m) - E_{2B}^*(m)}{w_{\cos}^+}\right)$

$E_{2B}^*(m) = E_{\text{asymp}}(\text{is_inv}(m)) \frac{\Psi_m}{\Psi_m + 4.2}$

$E_{\text{asymp}} = -18.2\ k_BT$ (open screw) vs. $-15.5\ k_BT$ (C_2 involution)

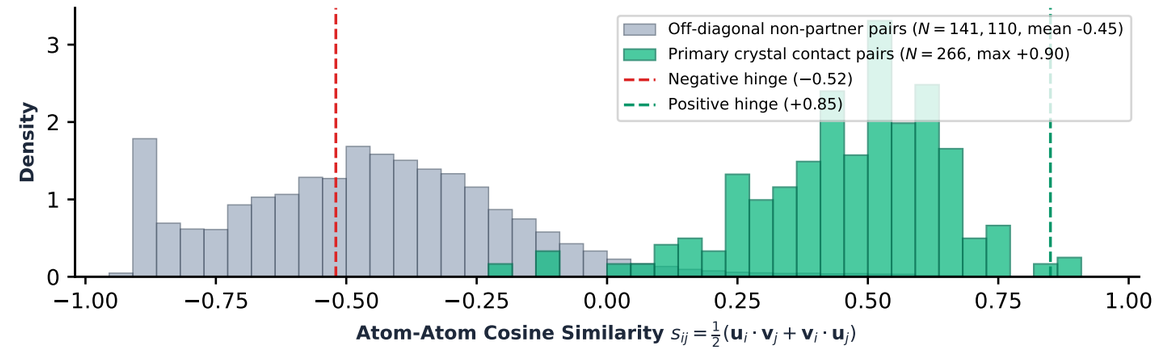
$$\mathcal{L}_{\text{Stage B}}(\mathbf{a}) = w_{\text{pose}} \sum_{m=1}^{8} \text{band}(S_{\cos}(x_m; \mathbf{a}),\ S_{\cos}^*(m)) + w_{\text{pert}} \sum_{p \in \mathcal{D}_{\text{pert}}} [S_{\cos}(x_p; \mathbf{a}) + 0.35]_+^2 + \lambda_a \|\mathbf{a} - a_{\text{bg}}\|^2$$

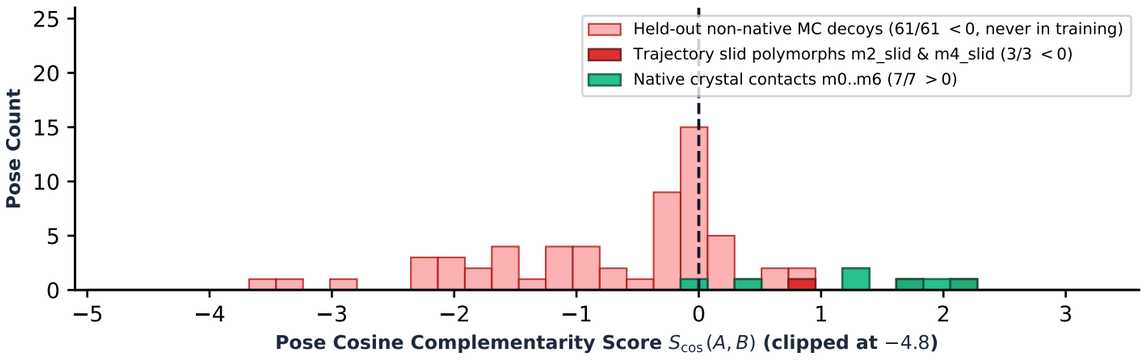
b Universal Stage B Energy-Balance Link Function $E_{2B}^*(\Psi_m)$



Polar Contact Support $\Psi_m = \sum_{(i,j) \in m} \psi(u_{ij})$

c Stage A Atom-Pair Partition Separation & Held-Out Pose Scores





d Unseeded 256-Monomer 3D $P4_32_12$ Nucleation (`nuc-55`, 10^6 Sweeps)

